BTS – IV – 04.13 – 0318

B. Tech. Degree IV Semester Examination April 2013

ME 404 APPLIED THERMODYNAMICS

(*2006 Scheme*)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 x 5 = 40)

I. (a) State and prove Carnot theorem.

(b) Briefly explain two statements of second law of thermodynamics.

(c) Describe briefly the mercury-water binary cycle.

(d) Draw T-S diagram of water and briefly explain steam generation process.

(e) Briefly explain Dalton's law and Gibb's Dalton law.

(f) Describe the working of axial flow compressor.

(g) How Junker's gas calorimeter is used to find the calorific value of fuels?

(h) Explain briefly the theoretical air and excess air for combustion.

PART B

(4 x 15 = 60)

II. (a) A reversible engine is supplied with heat from two constant temperature source 900K and 600K. The sink is at 300K. The engine develops work equivalent to 90KJ/S and reject heat at the rate of $56\frac{KJ}{S}$. Estimate (i) Heat supplied by each source (ii) Thermal efficiency of engine. (9)

(b) Derive Maxwell relations. (6)

OR

III. (a) Internal energy of a certain substance is given by the following equation (8)

$$u = 3.56\ pv_s + 84$$

where u in $\frac{KJ}{Kg}$, p in KPa and v_s in $\frac{m3}{Kg}$.

A system composed of 3 kg of this substance expands from an initial state of 500KPa and volume of $0.22m^3$ to a final state of 100KPa in a process in which pressure and volume are related by equation $pv_s^{1.2}$ = constant. If expansion is quasi-static, find $\theta, \Delta u$ and W for the process.

(b) Derive T ds equations. (7)

IV. (a) 1 kg of steam at a pressure 17.5 bar and dryness 0.95 is heated at constant pressure until it is completely dry. Determine (i) increase in volume (ii) quantity of heat supplied (iii) change in entropy. (7)

(b) Explain reheat cycle in steam power plant and its thermodynamic analysis with a flow diagram and T-S diagram. (8)

OR

V. (a) A steam power plant operating a simple ideal Rankine cycle. Steam enters the turbine at 3MPa and 400°C and is condensed in a condenser at a pressure of 80 kPa. Determine thermal efficiency, steam rate and heat rate. For a mass flow rate of 120kg/s of steam determine the power developed. (12)

(b) Draw T-S diagram of Rankine cycle when steam supplied at the inlet of turbine is (i) dry saturated (ii) superheated (iii) wet with a dryness fraction x. (3)

(P.T.O)

VI. Two vessels A and B both containing nitrogen are connected by a valve which is opened to allow the contents to mix and achieve an equilibrium temperature of 27°C. Before mixing the following information is known about the gases in the two vessels. (15)

Vessel A	Vessel B
P = 1.5 MPa	P = 0.6 MPa
t = 50^0C	t = 20^0C
Contents = 0.5 kg mol	Contents = 2.5 kg.

Calculate the final equilibrium pressure and amount of heat transferred to the surroundings. If the vessel had been perfectly insulated calculate the final temperature and pressure which would have been reached. Take r_{N2}=1.4.

OR

VII. (a) A three stage compressor compress air from 1 bar to 35 bar and delivers it at high pressure to receiver. The initial temperature is 17°C. Law of compression is $PV^{1.25}$ = constant is same for each stage. Assuming conditions of minimum work and take (CP) air = 1.005 $\frac{KJ}{KgK}$. (9)

(i) Find power required to deliver 14 m^3/min air measured at suction conditions.

(ii) Find the intermediate pressures

(iii) Find the amount of heat rejected in each intercooler.

(b) Derive the pressure ratio for minimum work requirement in a two stage reciprocating air compressor with perfect intercooling. (6)

VIII. (a) Ultimate analysis of dry coal burnt in a boiler by mass is (10)
C→ 84% H_2 → 9% and incombustibles → 7%.
Volumetric composition of dry fine gases is CO_2 → 8.75%, CO→ 2.25%, O_2→ 8%,N_2→81%. Calculate (i) minimum air required to burn 1 kg of coal (ii) mass of air actually supplied per kg of coal (iii) amount of excess air supplied per kg of coal burnt (iv) Air-fuel ratio.

(b) Briefly explain adiabatic flame temperature. (5)

OR

IX. (a) Apply first law of thermodynamics to combustion process. (9)

(b) Calculate ΔUo in $\frac{KJ}{Kg}$ for combustion of benzene (C_6H_6) vapour at 25°C. (6)

Given that $\Delta Ho = -3169100 \ \frac{KJ}{mole}$ and H_2O in vapour phase.

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